Digital Material Flow Intelligence: Comprehensive Data Sources and API Integration Strategies for the German Construction Waste (Bauschutt) Knowledge Graph

1. Executive Summary

The digital transformation of the German construction sector, driven by the imperatives of the Circular Economy (CE) and the European Green Deal, necessitates a fundamental restructuring of how material flows are documented, analyzed, and managed. Construction and demolition waste (*Bauschutt* and *Baumischabfall*) represents the single largest waste stream by mass in Germany, yet the data governing its lifecycle remains fragmented across static regulatory documents, disconnected silos of operational data, and emergent, disparate Application Programming Interfaces (APIs). The objective of this report is to provide an exhaustive, expert-level analysis of the data landscape required to populate a semantic Knowledge Graph dedicated to German construction waste management.

This report synthesizes technical specifications, regulatory frameworks, and physical material properties to address five critical dimensions: **Composition**, **Density**, **Contamination Indicators**, **Life Cycle Assessment (LCA) Data**, and **AVV (Abfallverzeichnis-Verordnung) Codes**. Our analysis reveals that constructing a robust Knowledge Graph requires the federation of three distinct data clusters: the **Regulatory Cluster** (AVV, LAGA M20, Ersatzbaustoffverordnung), the **Material Science Cluster** (LCA, Density, Crystallography), and the **Operational Cluster** (GISBAU, Municipal APIs, Material Passports).

We identify that while no single "God-API" exists to solve the construction waste challenge, high-quality data is available through specific endpoints such as the ÖKOBAUDAT API for environmental impacts, the Madaster API for material inventories, and the RegioIT administrative interfaces for disposal logistics. However, significant semantic gaps exist, particularly in the harmonization of "bulk density" (*Schüttdichte*) across different regional jurisdictions and the automated mapping of hazardous markers (GISCODEs) to waste classification codes. This report provides the blueprint for bridging these gaps, offering a detailed roadmap for data ingestion, ontology alignment, and the deployment of a queryable semantic web infrastructure that complies with the rigorous standards of the *Kreislaufwirtschaftsgesetz* (KrWG) and the *Mantelverordnung*.

2. Introduction: The Semantic Challenge of "Bauschutt" in the Information Age

2.1 The Context of Digital Circularity

The German construction industry stands at a precipice. With the enactment of the *Mantelverordnung* (MantelV) and the *Ersatzbaustoffverordnung* (EBV), the legal pressure to recycle mineral construction waste into high-quality secondary aggregates has never been higher. However, the physical reality of demolition waste is one of high entropy and low information density. A pile of rubble on a construction site differs fundamentally from a virgin product: it lacks a barcode, a standardized data sheet, or a clear chemical pedigree.

To treat *Bauschutt* as a resource ("Urban Mining"), we must re-attach data to matter. A Knowledge Graph (KG) serves as the ideal technological substrate for this task. Unlike rigid relational databases, a KG can model the flexible, hierarchical, and interconnected nature of waste data—linking a specific pile of concrete not just to its physical density, but to its regulatory status (AVV 17 01 01), its potential contamination history (e.g., built in 1970), and its carbon footprint (Scope 3 emissions).

2.2 The Problem of Heterogeneity

The primary obstacle to this digital integration is data heterogeneity. Information regarding construction waste is scattered across:

- **Legal Texts:** PDF documents defining pollutant limits (e.g., LAGA M20 [1]).
- **Scientific Databases:** APIs providing molecular weights and crystal structures (e.g., Materials Project [2]).
- **Logistics Tables:** Excel sheets used by local trucking companies to estimate tonnage from volume (e.g., Beton Eisack [3]).
- **Municipal Interfaces:** JSON endpoints for waste collection schedules (e.g., RegioIT [4]).

This report systematically deconstructs these sources, evaluating their fitness for programmatic ingestion into a Knowledge Graph. We distinguish between **Static Knowledge** (immutable constants like atomic weight or legal codes) and **Dynamic Intelligence** (variable data like current contamination levels or local disposal capacity).

2.3 Scope and Definitions

For the purposes of this report and the resulting Knowledge Graph, we strictly define the material scope based on the German AVV system:

- ***Bauschutt* (Mineral Waste):** Inert materials such as concrete, bricks, tiles, and ceramics (AVV 17 01 series). These materials are heavy, typically chemically stable, and candidates for high-grade recycling (RC-Beton).
- ***Baumischabfall* (Mixed Construction Waste):** A heterogeneous mixture containing

non-mineral fractions like wood, plastics, gypsum, and metals (AVV 17 09 04). This stream requires complex sorting and often has a lower density and higher calorific value.
- ***Kontaminierte Bausubstanz* (Contaminated Material):** Waste containing hazardous substances (AVV codes with an asterisk *), such as asbestos, coal tar, or PCBs.

3. The Regulatory Backbone: AVV Codes and Classification Logic

The foundational ontology of any German waste management system is the *Abfallverzeichnis-Verordnung* (AVV), which transposes the European Waste Catalogue (EWC) into national law. A Knowledge Graph must treat these codes not merely as text strings, but as the central nodes from which all regulatory obligations inherit.

3.1 Taxonomy of Construction Waste (Chapter 17)

The AVV is structured hierarchically, a feature that maps directly to the rdfs:subClassOf property in semantic web ontologies. Chapter 17 covers "Construction and Demolition Wastes (including excavated soil from contaminated sites)."

3.1.1 Hierarchical Structure and Graph Implications

The classification logic flows from the chapter (2-digit) to the sub-chapter (4-digit) to the specific waste code (6-digit).

- **Chapter 17:** Construction and Demolition Waste.
  - **Sub-Chapter 17 01:** Concrete, bricks, tiles, and ceramics.
    - **17 01 01:** Concrete. (The "clean" variant).
    - **17 01 06*:** Mixtures of, or separate fractions of concrete, bricks, tiles and ceramics containing hazardous substances. (The "dirty" variant).
  - **Sub-Chapter 17 09:** Other construction and demolition wastes.
    - **17 09 04:** Mixed construction and demolition wastes other than those mentioned in 17 09 01, 17 09 02 and 17 09 03.

Insight for the Knowledge Graph:
The distinction between "mirror entries" is critical. Code 17 01 01 (Concrete) and 17 01 06* (Hazardous Concrete) describe the same material base but different chemical states.
- **Graph Logic:** The Knowledge Graph should model a "Base Material" class (e.g., mat:Concrete) which can be instantiated as either avv:170101 or avv:170106_star depending on the value of a hasPollutantConcentration property. This allows the system to dynamically reclassify waste as new chemical analysis data (e.g., from LAGA testing) becomes available.[5]

3.1.2 Hazardous Waste Indicators ("Mirror Entries")

Wastes marked with an asterisk (*) are absolute hazardous entries or mirror entries triggered

by specific thresholds.

- **Absolute Hazardous:** e.g., 17 06 01* (Insulation materials containing asbestos). No testing is required to declare this hazardous; the presence of asbestos is sufficient.
- **Mirror Entries:** e.g., 17 01 06*. Requires a "Hazardous Property Assessment" (HP 1 to HP 15) as defined in the Waste Framework Directive.
- **Data Integration:** The graph must integrate the "List of Lists" logic [7] to map specific chemical constituents (e.g., CAS numbers for Lead, Cadmium) to the HP criteria that trigger the asterisked AVV code.

3.2 Data Sources for AVV and EWC Codes

To populate the AVV nodes, reliance on static PDF lists is insufficient for a dynamic system.

- **Destatis (Statistisches Bundesamt):**
  - **Format:** The Federal Statistical Office provides the waste balance sheets and AVV classifications in Excel and CSV formats.[8]
  - **Utility:** These datasets provide the authoritative German text and the hierarchical relationships. They also often include statistical "generation volumes" (Abfallaufkommen), allowing the graph to benchmark a specific site's waste generation against national averages.
  - **Integration:** Regular ingestion of the destatis_abfallbilanz.csv ensures the ontology remains synchronized with official federal statistics.
- **EPA and International Mappings:**
  - While the AVV is German, it is chemically grounded in international standards. The US EPA's "Consolidated List of Lists" [7] and "HazWaste" tools [11] provide a machine-readable cross-walk between chemical names, CAS numbers, and waste codes (F, K, P, U lists in the US, which have chemical analogs in the EU system).
  - **Graph Value:** Use these mappings to enrich the AVV nodes with specific chemical synonyms. If a user searches for "Trichlorophenol waste," the graph can traverse the chemical synonym links to suggest relevant hazardous waste codes.[12]
- **Commercial and Open Code APIs:**
  - **EWC Codes API:** Services like ewccodes.com [14] provide a RESTful API returning the full EWC list with metadata including hazardous (boolean) and density (approximate).
  - **GitHub Repositories:** Open-source datasets (e.g., pedropro/TACO or waste-datasets-review) often contain CSV dumps of waste taxonomies used for training computer vision models.[15] These can serve as training data for an AI layer of the Knowledge Graph that identifies waste types from site photos.

3.3 Semantic Mapping Strategy

The Knowledge Graph should utilize owl:sameAs or skos:exactMatch to align the German AVV codes with the European EWC codes and the internal schemas of disposal companies (which often use proprietary material codes).

Table 1: Key AVV Nodes for Bauschutt Integration

AVV Code	German Designation	Classification Logic	Graph Property Links
17 01 01	Beton	Inert, Recyclable	hasDensity: ~2.4t/m3, isRecyclable: True
17 01 02	Ziegel	Inert, Recyclable	hasDensity: ~1.8t/m3, color: Red/Orange
17 01 07	Gemische aus Beton, Ziegeln...	Mixed Mineral	requiresSorting: False, LulucfCategory: Mineral
17 09 04	Gemischte Bau- und Abbruchabfälle	Mixed, Non-Hazardous	hasDensity: ~0.2-0.8t/m3, requiresSorting: True
17 01 06*	Gemische... mit gefährlichen Stoffen	**Hazardous**	requiresManifest: True, disposalPath: SpecialTreatment
17 06 05*	Asbesthaltige Baustoffe	**Hazardous**	hazardType: Carcinogenic, handling: TRGS519

4. Physical Material Intelligence: Composition and Density

A Knowledge Graph for waste management must bridge the gap between "volume" (spatial data from BIM/CAD) and "mass" (logistics data for transport and carbon accounting).

4.1 The Physics of Waste Density

Construction materials undergo a fundamental physical transformation when they become

waste. A solid concrete wall has a *Solid Density* (*Rohdichte*), typically ~2300-2400 kg/m³. When demolished, it becomes rubble with air voids, characterized by *Bulk Density* (*Schüttdichte*).

Critical Insight: Using solid density values for waste logistics results in massive overestimation of weight and transport requirements. The Knowledge Graph must explicitly distinguish between mat:solidDensity and mat:bulkDensity.

4.1.1 Granular Density Data Sources

Generic databases often fail to capture the nuance of "crushed" vs. "broken" vs. "mixed" waste. We identify specific, high-fidelity sources for this data:

- **Beton Eisack (Regional Processor Data):**
  - Operational tables from recycling centers [3] provide the most accurate "real-world" densities because they are based on weighbridge data.
  - **Data Points:**
    - *RC-Sand (0-10 mm):* 1.45 t/m³ (Higher density due to packing).
    - *RC-Kies (30-80 mm):* 1.38 t/m³ (Lower density due to large air voids).
    - *Schottergemisch (0-31.5 mm):* 1.64 t/m³ (Highest density, widely used for road bases).
  - **Graph Integration:** These values should be stored as regional instances (e.g., ex:Density_SouthTyrol_2021) to account for geological variations in aggregate density.
- **City of Vienna (MA 22) Guidelines:**
  - These guidelines [17] provide conversion factors specifically for the *disposal* phase, highlighting the extreme lightness of mixed waste.
  - **Data Points:**
    - *Baustellenabfälle (gemischt):* **0.20 t/m³**. This extremely low figure reflects the high volume of insulation, packaging, and plastics in "Baumischabfall" compared to pure mineral "Bauschutt."
    - *Bodenaushub (Excavated Soil):* 1.50 t/m³.
  - **Implication:** A container of "mixed construction waste" weighs ~85% less than a container of "concrete rubble." The Knowledge Graph must use the AVV classification to select the correct density factor automatically.
- **Canton Uri (Switzerland) Guidelines:**
  - Detailed factors for rail track ballast and excavated materials.[18]
  - *Betongranulat:* 1.7 - 1.9 t/m³ (Note: Higher than Beton Eisack, likely referring to compacted state).
  - *Asphaltgranulat:* 1.6 - 1.8 t/m³.

4.1.2 Scientific and Generic APIs

For theoretical baselines and pure material properties, scientific APIs are valuable.

- **Materials Project API:** [2]
  - **Endpoint:** https://api.materialsproject.org/
  - **Data:** Crystallographic density of pure compounds (e.g., SiO2, CaCO3).
  - **Relevance:** Useful for "first principles" modeling or when analyzing novel materials where empirical waste data is missing. The API requires an API key and supports extensive querying via Python (mp-api).

4.2 Composition and Material Identity

Before a material becomes waste, it is a product. Connecting the "End of Life" graph to the "Design" graph is achieved through Material Passports.

4.2.1 Madaster (Material Cadastre)

Madaster [19] is a platform designed to register materials within buildings, effectively creating a "bank" of materials.

- **API Architecture:**
  - **Endpoint:** https://api.madaster.com/v4.0/
  - **Authentication:** X-API-Key header.
  - **Core Entities:** Building, Material, Passport.
- **Material Families (V2):**
  - Madaster categorizes products into families (e.g., "I - Wood") and sub-families ("I1 - Timber").
  - **Graph Strategy:** When a building in the Knowledge Graph is flagged as "To Be Demolished," the system should query the Madaster API to retrieve the BillOfMaterials. This JSON object provides the exact *composition* of the waste stream before the excavator even arrives.
- **Quantities:** The API provides precise volume, area, and weight data derived from the source IFC/BIM files. This eliminates the need for estimation if the building has a digital twin.

4.2.2 2050 Materials

- **Platform:** [22] A database of sustainable building products.
- **Taxonomy:** Uses Uniclass and CSI Masterformat.
- **API Utility:** Can be used to resolve generic terms (e.g., "Insulation") into specific product types with known chemical compositions, improving the accuracy of the contamination risk assessment.

Table 2: Density Data Sources Comparison

Source	Material Context	Density Type	Typical Value (Concrete)	API/Format

Beton Eisack	Recycling Center (Crushed)	Bulk (Schütt)	~1.50 t/m³	PDF Table
City of Vienna	Disposal Logistics (Mixed)	Bulk (Loose)	0.20 t/m³ (Mixed)	PDF Guidelines
Canton Uri	Road Construction (Compacted)	Bulk (Compacted)	1.7 - 1.9 t/m³	PDF Table
Materials Project	Material Science (Pure)	Solid (Crystal)	~2.40 t/m³	REST API (JSON)
Madaster	BIM/Inventory	Product Specific	Variable	REST API (JSON)

5. Chemical Intelligence: Contamination Indicators and Hazards

The transition from "Product" to "Waste" is often triggered by contamination. A key function of the Knowledge Graph is to flag materials that *cannot* be recycled due to chemical pollutants (Scope of LAGA M20/EBV).

5.1 The "Schadstoff" Challenge

Legacy buildings act as reservoirs for substances now banned. The graph must link "Building Era" to "Pollutant Probability."

- **Asbestos:** Common in fiber cement, floor adhesives, and fireproofing (pre-1993).
- **PCB (Polychlorinated Biphenyls):** Joint sealants, paints, capacitors (pre-1980s).
- **PAK (Polycyclic Aromatic Hydrocarbons):** Tar-based adhesives, roofing felt, asphalt flooring.
- **Heavy Metals:** Lead pipes, anti-corrosion paints.

5.2 Regulatory Limits: LAGA M20 and Ersatzbaustoffverordnung

The classification of mineral waste for reuse is governed by strict limit values.

- **LAGA M20:** The traditional guidelines. Defines classes Z0 (Unrestricted), Z1.1/Z1.2 (Restricted), and Z2 (Technical safety measures required).

- **Ersatzbaustoffverordnung (EBV):** The new federal ordinance (part of MantelV). Defines material classes like *RC-1*, *RC-2*, *RC-3* based on strict eluate and solid content limits.

Data Integration:

- **Threshold Tables:** The graph must contain the limit value tables for both LAGA and EBV.[1]
  - *Example Logic:* IF Sulphate > 50 mg/l THEN Class!= RC-1.
  - *Source:* Open-source repositories (e.g., "ZOW-Tabelle-Boden-EBV" on GitHub [25]) often contain these limit values in JSON or JavaScript arrays, ready for ingestion.
- **Parameters:** Key indicators include Conductivity, pH, TOC (Total Organic Carbon), Arsenic, Lead, Cadmium, Chromium, Copper, Nickel, Mercury, Zinc, EOX, and PAK.

5.3 Operational Hazard Data: GISBAU and WINGIS

For chemical products used in construction (and appearing as waste residues), **GISBAU** (Gefahrstoff-Informationssystem der BG BAU) is the authoritative source.[26]

- **GISCODES:** A simplified coding system (e.g., "RE1" for epoxy resins, "ZP1" for cement products) that groups products with similar hazards.
- **Integration:**
  - The Knowledge Graph should map GISCODES to AVV codes.
  - *Mapping Example:* A floor covering with adhesive GISCODE S1 (containing solvents) should trigger a flag for AVV 17 09 03* (Construction waste containing hazardous substances) if not separated.
- **Access:** WINGIS online provides a searchable interface. While a public REST API is not explicitly documented, the underlying data structure links trade names to hazard markers, which can be scraped or requested for the graph.

5.4 Institutional Disposal Guidelines

Universities often publish detailed "Abfall-ABC" (Waste ABC) guidelines which serve as excellent proxies for detailed sorting logic.[29]

- **Chemical Bottles:**
  - *Standard:* Glass bottles must be "drip-free," rinsed, and labels neutralized to be recycled as glass.
  - *Hazard:* If residues remain, they are "Hazardous Waste" (Sonderabfall) and must be logged via specific forms.
- **Graph Logic:** The graph can incorporate these "procedural rules" as validation steps. e.g., Node:ChemicalBottle -> Action:CheckResidue -> Decision:Recycle(Glass) OR Dispose(Hazard).

6. Environmental Intelligence: LCA and Sustainability Data

To support the Circular Economy, the Knowledge Graph must calculate the environmental cost of disposal decisions. This requires linking waste mass to Life Cycle Assessment (LCA) datasets.

6.1 ÖKOBAUDAT: The Primary Source

ÖKOBAUDAT is the official German platform for LCA data, compliant with EN 15804.[32]

- **API & Format:**
  - **Endpoint:** https://oekobaudat.de/OEKOBAUDAT/resource
  - **Format:** ILCD (International Life Cycle Data System) XML and JSON.
  - **Content:** Contains datasets for construction products with modules A1-A3 (Production), C3 (Recycling), C4 (Landfill), and D (Benefits/Loads).
- **Relevance to Waste:**
  - The database explicitly models **End-of-Life (EoL)** scenarios.
  - *Example:* A dataset for "Concrete C30/37" will have specific GWP (Global Warming Potential) values for "Scenario: Recycling" vs. "Scenario: Landfill."
- **Data Quality Insights:**
  - **Clumping:** Analysis of Ökobaudat data [32] reveals data points often "clump" around generic values (e.g., 230 kg CO2/m³ for concrete). The Knowledge Graph should prioritize manufacturer-specific EPDs where available to avoid generic averages.
  - **Missing Modules:** Some datasets lack Module C2 (Transport to disposal) or A4 (Transport to site). The Knowledge Graph needs a separate "Transport Logic" layer (calculating Distance * Mass * Truck_Emission_Factor) to fill these gaps.

6.2 OpenEPD and Building Transparency

- **EC3 API:** [35]
  - **Access:** Requires an account and token. Throttled at 45 tokens/minute.
  - **Format:** OpenEPD (JSON).
  - **Value:** Access to a global database of Environmental Product Declarations (EPDs).
- **Graph Strategy:** Use EC3 to fetch "Embodied Carbon" data for materials entering the waste stream. This allows the graph to calculate the "Lost Carbon" if a material is landfilled instead of reused.

6.3 Scope 3 Calculation Logic

The Knowledge Graph integrates these sources to perform the calculation:

$$\text{Total Impact} = (\text{Mass} \times \text{EoL Factor}_{\text{ÖKO}}) + (\text{Mass} \times \text{Distance} \times \text{Transport Factor}) - (\text{Mass} \times \text{Recycling Credit}_{\text{Module D}})$$

This formula allows the system to output a "Net Environmental Impact" for every ton of

Bauschutt managed.

7. Operational Intelligence: Logistics and Disposal

The final layer of the graph connects "Material" to "Location." Waste needs a destination.

7.1 Municipal APIs: RegioIT / Abfallnavi

For waste generated in urban environments (municipal waste, small construction sites), the **RegioIT Abfall API** [4] is a key infrastructure component.

- **Infrastructure:** Used by many German municipalities (Aachen, Nürnberg, Gütersloh, etc.).
- **API Structure:**
  - **Base URL:** https://{region}-abfallapp.regioit.de/abfall-app-{region}/rest
  - **Endpoints:**
    - /fraktionen: Returns the list of waste types (Bio, Paper, Restmüll, Bauschutt). This connects the "Concept" of waste to the "Local Definition."
    - /standorte: Returns geolocation (lat/long) and opening hours of recycling centers (Recyclinghöfe).
    - /termine: Collection dates.
- **Swagger/OpenAPI:** The API definition is often available via Swagger UI [4], enabling automated client generation for the Knowledge Graph.
- **Graph Integration:**
  - Map avv:170101 to the specific fraktionId used by the local RegioIT instance.
  - Use /standorte to populate the ex:DisposalFacility nodes in the graph, calculating the Euclidean distance from the demolition site to the facility to optimize transport emissions.

7.2 Disposal Facilities and Logistics

- **BSR (Berliner Stadtreinigung):** [38]
  - Provides "Abfall-ABC" data and facility locations in JSON format via Berlin Open Data.
  - **Buf Schema Registry:** References to BSR APIs using Protobuf/gRPC suggest a modern, high-performance interface for high-volume data exchange.[40]
- **Container Services:**
  - Private waste management companies (e.g., REMEX, Veolia) often have customer portals. While public APIs are rarer, their "Acceptance Catalogues" (based on AVV) can be scraped or ingested as CSVs to define ex:acceptsWasteCode properties for facility nodes.

8. Technical Architecture and Integration Strategy

Synthesizing these diverse sources requires a disciplined ETL (Extract, Transform, Load)

pipeline.

8.1 Ontology Design

The ontology should be built around three core classes:

1. **MaterialBatch:** Represents the physical waste.
    - Properties: hasMass (derived from Density), hasVolume (from Madaster), hasContaminantLevel (from Lab Analysis).
2. **WasteClassification:** Represents the regulatory bucket.
    - Properties: hasAVVCode, isHazardous, requiresManifest.
3. **ProcessingOption:** Represents the destination.
    - Properties: location, acceptsAVV, hasGWPImpact.

8.2 Data Ingestion Pipeline

1. **Ingest Static Regulatory Data:**
    - Download **AVV** and **LAGA M20** tables (CSV/JSON).
    - Map LAGA limits to sh:SHACL shapes (validation rules) in the graph. *Rule: If Lead > X, invalidate Z0 classification*.
2. **Ingest Material Science Data:**
    - Query **ÖKOBAUDAT** (XML) for all UUIDs related to "Concrete", "Bricks", "Ceramics". Extract density and GWP values.
    - Ingest **Beton Eisack** density tables to create a "Regional Density Map."
3. **Connect Operational Data:**
    - Build adapters for **Madaster API** to pull demolition inventories.
    - Build adapters for **RegioIT API** to fetch valid disposal sites.

8.3 Handling Data Quality and Uncertainty

- **Density Ranges:** Since waste density varies (e.g., 1.4 - 1.6 t/m^3), the graph should store minDensity, maxDensity, and probabilisticMean.
- **Clumping:** Flag ÖKOBAUDAT datasets that use generic averages with a dataQuality:Low marker, preferring specific EPDs where available.

9. Conclusion

The construction of a German Waste Management Knowledge Graph is not constrained by a lack of data, but by the need for semantic harmonization. We have identified a rich ecosystem of data sources:

- **Regulatory:** The AVV and LAGA/EBV provide the "Rules of the Game," available in structured formats from Destatis and open-source repos.
- **Physical:** Regional tables (Beton Eisack, Vienna) provide the "Ground Truth" for density, correcting the theoretical bias of scientific databases (Materials Project).
- **Environmental:** ÖKOBAUDAT and OpenEPD provide the "Impact Metrics," essential for

the Circular Economy.
- **Operational:** Madaster and RegioIT APIs provide the "Live Status" of materials and facilities.

By federating these sources, the Knowledge Graph can answer complex queries such as *"Find a disposal facility within 20km for 500 tons of slightly contaminated concrete (AVV 17 01 01, Sulfate < 100mg/l) and calculate the transport CO2."* This capability is the prerequisite for a truly circular construction sector in Germany.

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[1]

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